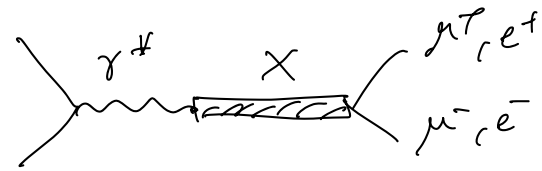


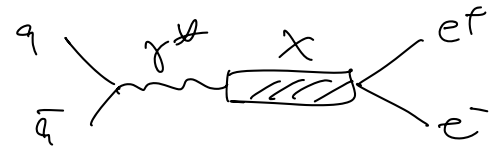
J/ψ charmonium (c c̄)

① $e^+e^- \rightarrow c\bar{c} \rightarrow \mu^+\mu^-, e^+e^-$



② $p + Be \rightarrow (c\bar{c}) + X$

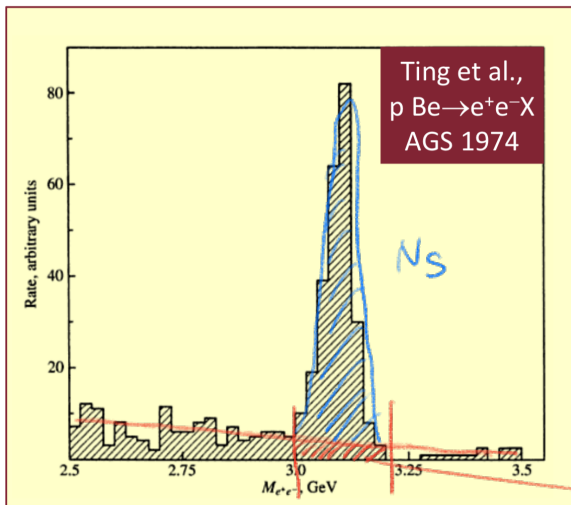
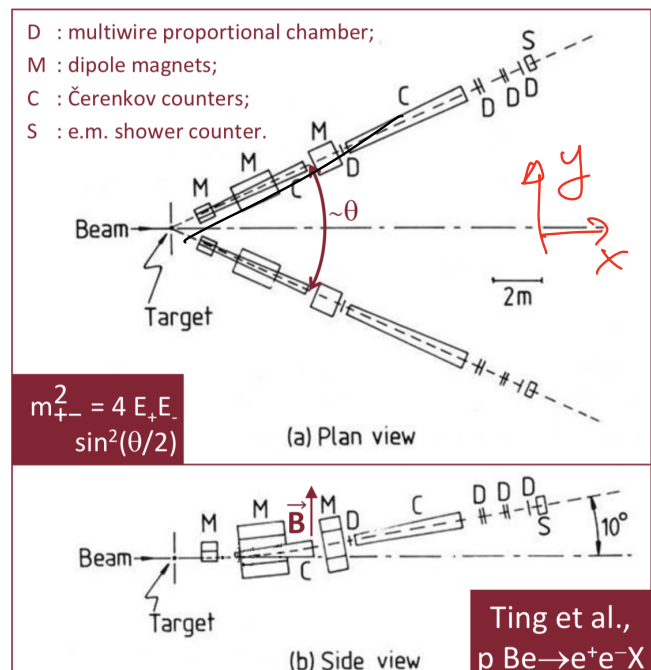
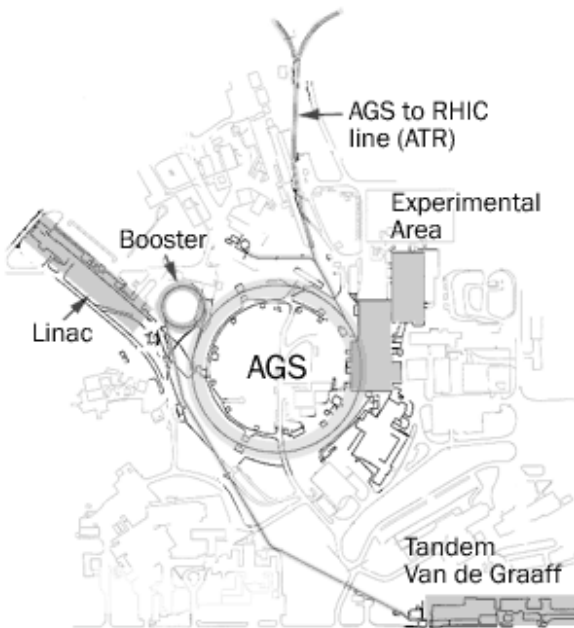
$\hookrightarrow e^+e^-, \mu^+\mu^-$



mass resolution better for e^+e^-

$\Rightarrow$ look for a peak in invariant mass M_{ee}

hadronic mode



$\sigma_{ee} \sim 20 \text{ MeV}$

peak $M_{ee} \sim 3.1 \text{ GeV}$

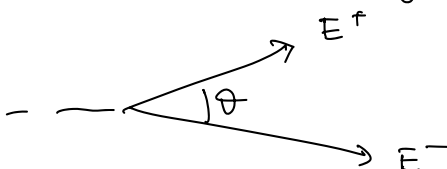
$$\frac{N_S}{\sqrt{N_B}} = \frac{N_{tot} - N_B}{\sqrt{N_B}} = \text{significance}$$

$N_B \approx 40 \text{ events}$

$N_{tot} = 250$

$\Rightarrow \text{signif} \approx \frac{250 - 40}{\sqrt{40}} = \frac{210}{6.5}$

Sam Ting



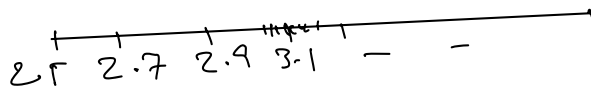
$$M_{ee} = 4 E^+ E^- \sin^2 \frac{\theta}{2}$$

you measure E^+, E^-, θ

SLAC: $e^+e^- \rightarrow X \rightarrow e^+e^-, \mu^+\mu^-, \text{hadrons}$.

SPEAR: Super positron electron asymm. ring.

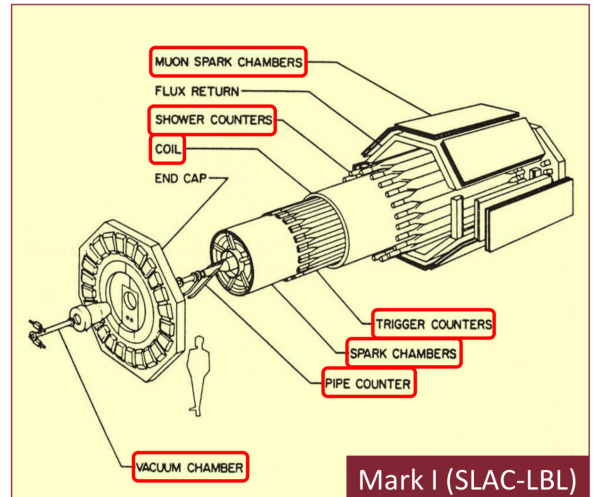
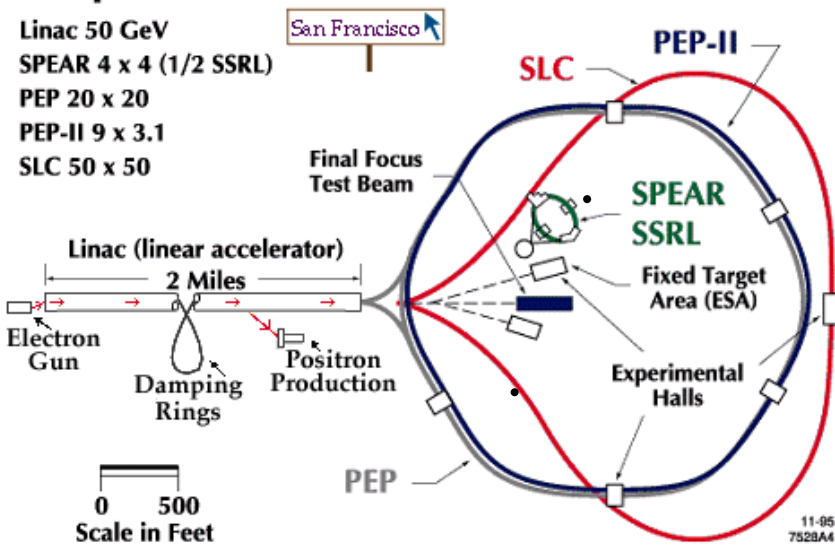
Scan of $\sqrt{s}$: $2.5 \rightarrow 7.5$ GeV steps of 200 MeV



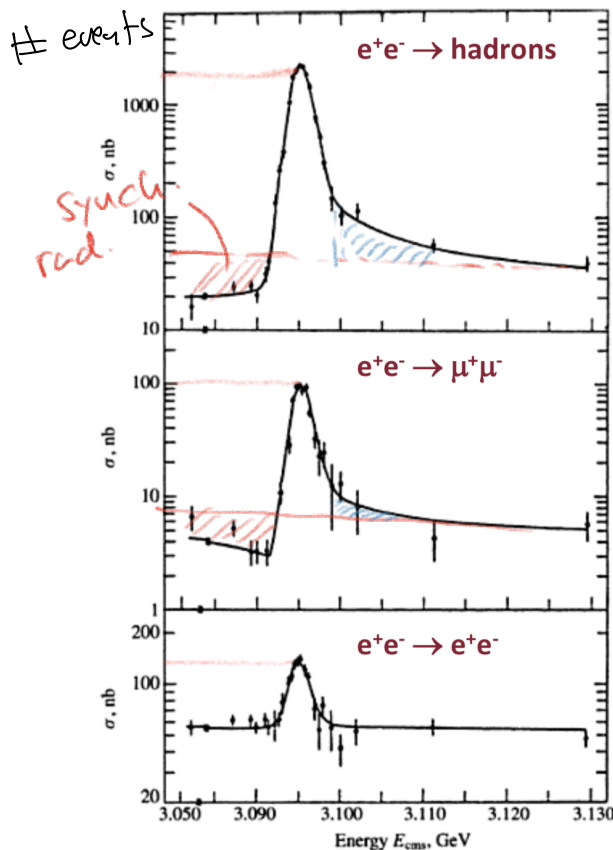
Scan around 3.1 GeV with step of 2.5 MeV

Experimental Areas at SLAC

Linac 50 GeV
SPEAR 4 x 4 (1/2 SSRL)
PEP 20 x 20
PEP-II 9 x 3.1
SLC 50 x 50



Mark I (SLAC-LBL)



Measure # events as function of $\sqrt{s}$

$$\sqrt{s} = 2E_{\text{beam}}$$

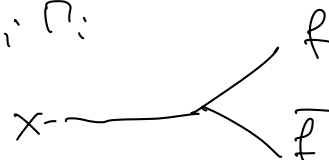
$$\sigma(e^+e^- \rightarrow X \rightarrow f\bar{f}) =$$

$$\frac{16\pi}{s} \frac{2J_X + 1}{(2S_1 + 1)(2S_2 + 1)} \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{f\bar{f}}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\text{tot}}^2}{\Gamma_{\text{tot}}^2 + (m_X - \sqrt{s})^2}$$

Γ_{tot} : total decay width of X

$$B(X \rightarrow e^+e^-) = \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}}$$

$$= \frac{\Gamma(X \rightarrow ee)}{\sum_i \Gamma_i}$$



Burt Richter @ SLAC

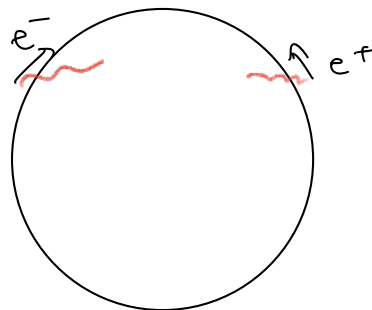
Peak of σ for $\sqrt{s} = m_X$

$$\sqrt{s} = E_1 + E_2 = E^+ + E^-$$

Nominal $\sqrt{s} = E^+ + E^-$

$e^\pm$ can emit γ before collision.

$$\sqrt{s}_{\text{eff}} = E^+ + E^- - E_\gamma < \sqrt{s}_{\text{nominal}}$$



synchrotron radiation for $e^\pm$ before collision

$\sqrt{s}_{\text{nom}} = 3050 \text{ MeV}$ below the peak.

$\sqrt{s}_{\text{eff}} < 3050 \text{ MeV}$ because of synch. rad.

$\sigma_{\text{eff}} < \sigma_{\text{nominal}}$ @ 3050 MeV

$\sqrt{s}_{\text{nom}} = 3150 \text{ MeV} \Rightarrow \sigma(3150) < \sigma(3100)$ because above peak.

because of synch. rad. $\sqrt{s}_{\text{nom}} = 3150 \rightarrow \sqrt{s}_{\text{eff}} = 3100$

$\Rightarrow$ more events than nominal cross section.

Resonance supposed to be $(c\bar{c})$ very narrow.

Observed width of the peak $\sim$ detector resolution



$\Rightarrow \Gamma_{c\bar{c}}$ very small.

$$e^+e^- \rightarrow \gamma^* \rightarrow X \rightarrow e^+e^-$$

X should have same quantum numbers as γ

Tiny: J Richter: $\psi \Rightarrow J/\psi$ in PDG

$$c\bar{c} \text{ } J/\psi: \quad J^{PC} = 1^{--} \quad P = (-1)^{L+1}$$

$L=0$ for ground state

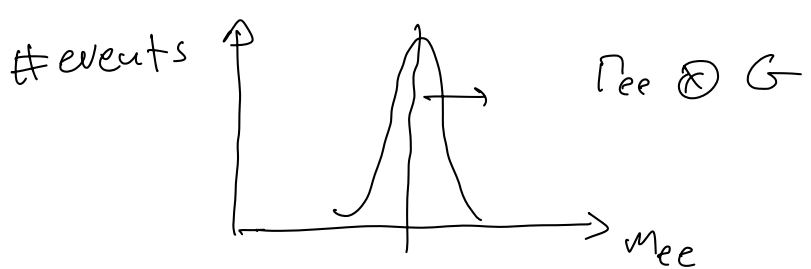
$$C = (-1)^{L+S} \quad L=0 \Rightarrow S=1$$

J/ψ is a vector meson.

J/ψ width

$$J^{PC} = 1^{--}$$

$$\frac{\Gamma}{M} \ll 1 \quad \text{narrow resource}$$



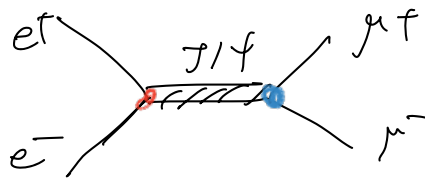
$\Gamma = \sum_i \Gamma_i = \Gamma_{tot}$ of the particle
In e^+e^- experiment:

$$\#(e^+e^- \rightarrow had) \propto \sigma(e^+e^- \rightarrow had) \propto$$

$$\frac{\Gamma_{ee}}{\Gamma_{tot}} \quad \frac{\Gamma_{had}}{\Gamma_{tot}} \quad \text{had BF.}$$

$$\#(e^+e^- \rightarrow \mu^+\mu^-) \propto \sigma(e^+e^- \rightarrow \mu^+\mu^-) \propto$$

$$\frac{\Gamma_{ee}}{\Gamma_{tot}} \quad \frac{\Gamma_{\mu\mu}}{\Gamma_{tot}} \quad \text{prod.} \quad \text{delay.}$$



$$\#(e^+e^- \rightarrow e^+e^-) \propto \sigma(e^+e^- \rightarrow e^+e^-) \propto \frac{\Gamma_{ee}}{\Gamma_{tot}} \quad \frac{\Gamma_{ee}}{\Gamma_{tot}}$$

4 unknowns: $\Gamma_{ee}, \Gamma_{\mu\mu}, \Gamma_{had}, \Gamma_{tot}$

3 observables: $\sigma_{ee}, \sigma_{\mu\mu}, \sigma_{had}$. (measured from the peak)

1 constraint: $\Gamma_{tot} = \Gamma_{ee} + \Gamma_{\mu\mu} + \Gamma_{had}$

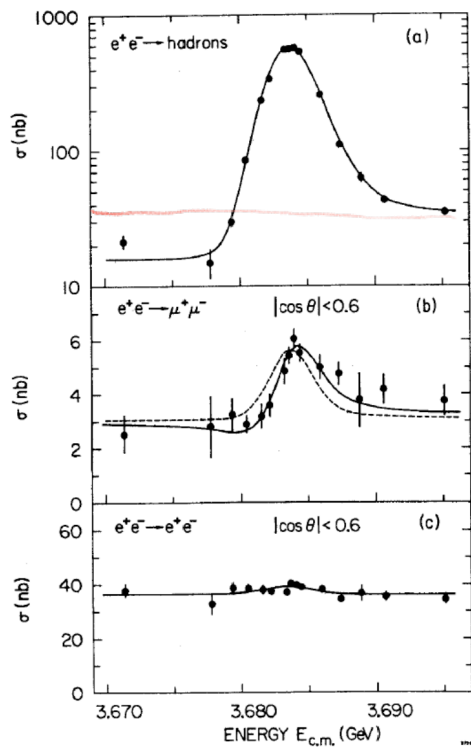
$$\Rightarrow \Gamma_{tot} = 0.087 \text{ MeV} = 87 \text{ KeV}$$

$$\rho(770) \quad m = 770 \text{ MeV} \quad \Gamma = 150 \text{ MeV} \quad \frac{u\bar{u} - d\bar{d}}{\sqrt{2}} \quad J^{PC} = 1^{--}$$

$$\phi \quad m = 1020 \text{ MeV} \quad \Gamma = 4 \text{ MeV} \quad s\bar{s} \quad J^{PC} = 1^{--}$$

$$J/\psi \quad m = 3.1 \text{ GeV} \quad \Gamma = 87 \text{ KeV} \quad c\bar{c} \quad J^{PC} = 1^{--}$$

$c\bar{c}$ is a resonance $\Rightarrow$ there should be other excited $c\bar{c}$ states.



10 days later @ SPEAR

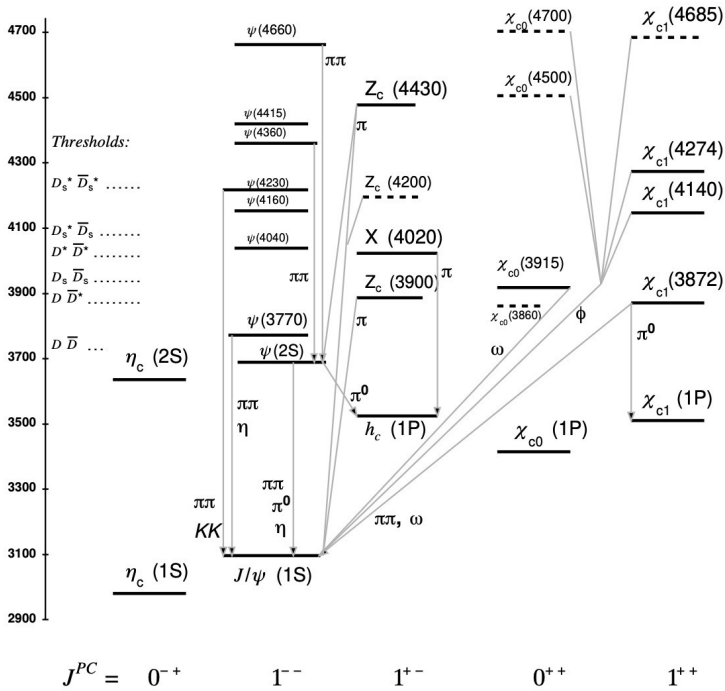
$$\sqrt{s} \approx 3.68 \text{ GeV} > m_{J/\psi}$$

$\psi(2S)$

$$m = 3.7 \text{ GeV}$$

$$\Gamma = 294 \text{ KeV}$$

Mass (MeV)



Charmonium Resonances $c\bar{c}$

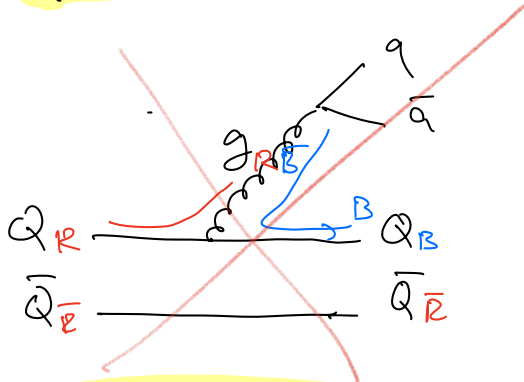
$\chi_{c2}(3930)$ $\psi_2(3823)$ $\psi_3(3842)$
 $\chi_{c2}(1P)$ $\chi_{c1}(1P)$

Width of $Q\bar{Q}$ resonances

Difference between ρ, ϕ, Υ

$Q\bar{Q}$ $Q = u, d, s, c$

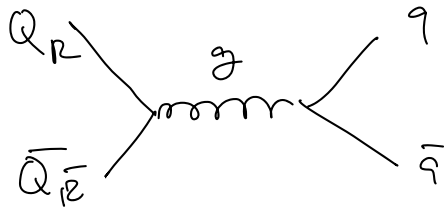
Spectator diagram



No color-less gluon.

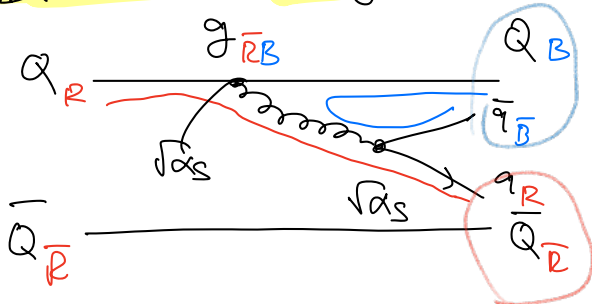
$\Rightarrow$ hadrons would be colored
Not possible

Anihilation diagram (s-channel)



Same problem $\Rightarrow$ not possible.
Hadrons must be colorless

Internal diagram



$M \sim \Lambda_s$

$X \rightarrow \text{had}_1 + \text{had}_2$

$\rho(770)$

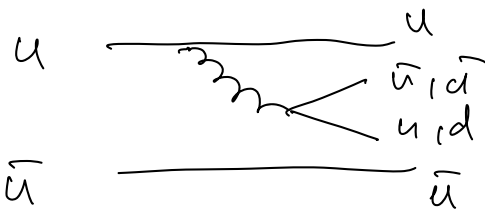
$\Gamma = 150 \text{ MeV}$

$M = 770 >> 2 \times m_\pi = 280 \text{ MeV}$

$$\rho = \frac{u\bar{u} - d\bar{d}}{\sqrt{2}}$$

$$\rho \rightarrow \pi^+ \pi^-$$

$$\pi^0 \pi^0$$



$\rho \rightarrow K^+ K^-$ $Q < 0$
Kinematically not allowed.

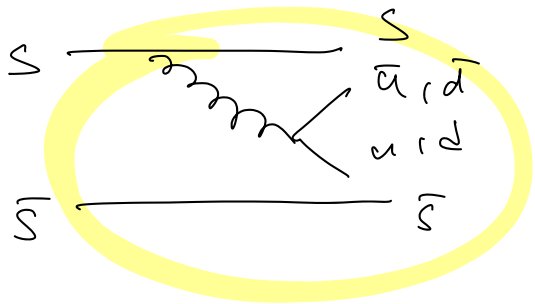
$\rho(770)$ DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1 $\pi\pi$	~ 100	%
Γ_2 $K\bar{K}$		

$\phi(1020)$

$$\phi \rightarrow K^+ K^-$$

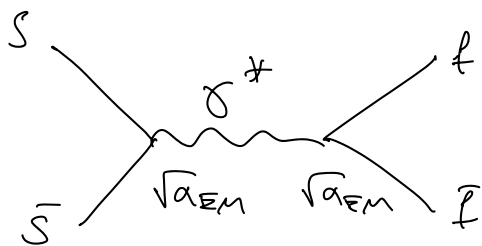
$$K^0 \bar{K}^0$$



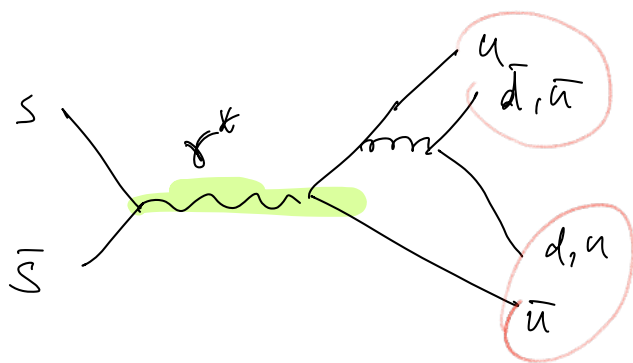
$\phi(1020)$ DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1 $K^+ K^-$	$(49.1 \pm 0.5) \%$	$S=1.3$
Γ_2 $K_L^0 K_S^0$	$(33.9 \pm 0.4) \%$	$S=1.2$
Γ_3 $\rho\pi + \pi^+\pi^-\pi^0$	$(15.4 \pm 0.4) \%$	$S=1.2$
Γ_4 $\rho\pi$		
Γ_5 $\pi^+\pi^-\pi^0$		
Γ_6 $\eta\gamma$	$(1.301 \pm 0.025) \%$	$S=1.2$
Γ_7 $\pi^0\gamma$	$(1.32 \pm 0.05) \times 10^{-3}$	
Γ_8 $\ell^+\ell^-$	—	
Γ_9 e^+e^-	$(2.979 \pm 0.033) \times 10^{-4}$	$S=1.3$
Γ_{10} $\mu^+\mu^-$	$(2.85 \pm 0.19) \times 10^{-4}$	

$$\alpha_s \approx 0.1$$



EM decay of ϕ $\alpha_{EM} \sim \frac{1}{137}$



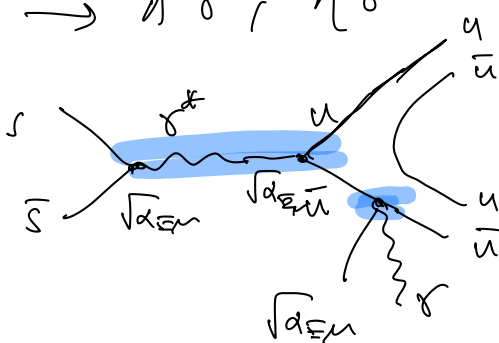
$$\phi \rightarrow \pi^+ \pi^-$$

$$\pi^0 \pi^0$$

$$\phi \rightarrow \pi \rho$$

$$J \quad 1 \quad 0 \quad 1$$

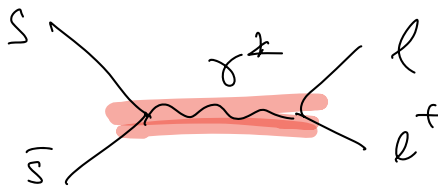
$$\phi \rightarrow \pi^0 \sigma, \eta \sigma$$



$$\mathcal{M} \sim (\alpha_{EM})^{3/2}$$

$$\Rightarrow \text{BF}(\phi \rightarrow \pi^0 \sigma) \sim \alpha_{EM}^3$$

$$\phi \rightarrow \ell^+ \ell^- \text{ allowed}$$



No direct measurement

Phy Rev. Lett 86, 1698 (2001)

$$e^+ e^- \rightarrow \phi \rightarrow K K$$

$$\sqrt{s} = 1020 \text{ MeV}$$